

Training-Design Sensitivity of Structural-Prior Gains in Unrolled Channel Estimation for Rydberg Atomic Receivers

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Abstract—Adding a structural prior inside an unrolled channel estimator usually improves the reported metric, and the improvement is usually credited to the prior. We show that in magnitude-only channel estimation for Rydberg atomic receivers that credit is not established by an uncontrolled comparison, because the measured advantage depends strongly on how the two arms were trained. A per-sample normalized error averaged over a wide SNR range is dominated by its low-SNR tail [1]; here 89.7% of the measured gradient norm falls below 5 dB, so any mechanism that helps at high SNR can appear to supply information. We propose a matched-adequacy control: measure the per-bin gradient share, retrain both the structured and the unstructured arm with the imbalance removed, and re-measure. Applied to our own Hankel-structured estimator over three training designs with three seeds each, the structural advantage at $\text{SNR} \geq 5$ dB is +1.286 dB under the conventional loss, near zero on average under focused retraining, and negative in the balanced-loss runs. Analyzed as paired per-seed differences, the change from conventional training is -1.27 dB under focused retraining and -1.46 dB under the balanced loss, both with intervals excluding zero — while the individual focused and balanced contrasts are not themselves separated from zero on three seeds. A per-bin sign pattern survives where no scalar does. Correcting the loss is an adaptation of a published remedy, and we measure what it is worth here: +2.171 dB at $\text{SNR} \geq 5$ dB over three seeds, improving all eighteen bin-by-seed point estimates including the two bins the weighting penalizes most. A log-domain loss needing no bin edges does not resolve a difference from per-bin reweighting. Separately, pilot efficiency and pilot-count generalization differ by 2.231 dB in this setting. Results are simulation only.

Index Terms—Algorithm unrolling, channel estimation, evaluation methodology, phase retrieval, Rydberg atomic receiver, structural priors.

I. INTRODUCTION

UNROLLING an iterative estimator into a trained network [2], [3] has become standard for channel estimation, including for Rydberg atomic receivers, where the magnitude-only readout makes the problem one of phase retrieval [4], [5]. A recurring next step is to add an explicit structural prior — a low-rank projection, a sparsity constraint — inside the unrolled loop, and to report the resulting improvement as evidence that the prior supplies information the network lacks.

This paper argues that such evidence is usually not decisive, and that the reason is a convention so widespread it is rarely

stated: the training loss is a per-sample normalized error, averaged over a training set spanning a wide SNR range.

The estimator evaluated here is our own: we built a Hankel-structured unrolled network and measured an advantage at high SNR. The control in Sec. III was run to characterize that advantage; it removed it, and we report the result on our own method for the same reason we would report it on anyone's.

Related work on SNR-dependent loss design. The imbalance itself is known and we do not claim it. Wiesmayr *et al.* [1] state it directly for ML-assisted detection and decoding: a system learns one parameter set while operating over a range of SNRs, the training data is sampled from that range, and the aggregate loss is then dominated by the low-SNR samples, because a small relative improvement at low SNR moves the cost further than a large one at high SNR. They propose SNR deweighting as the remedy. They also observe, in their introduction, that systems are usually trained either at a single SNR or on samples drawn uniformly across the target range, apparently without the effect on performance at different SNRs being questioned — a published statement that the convention is unexamined, which is the gap this paper works in. Work on deep OFDM channel estimation has separately examined training on SNR-restricted samples, reporting the expected trade of low-noise accuracy against high-noise accuracy [6]. What we take from that line of work is the mechanism and the remedy. What is not taken from it, and what this paper is about, is the consequence for *attribution*: that a known training defect manufactures apparent evidence for a structural prior, so that crediting the prior without controlling for the defect is unsound. Wiesmayr *et al.* identify the pathology and fix it; they do not ask what it does to the evidence.

Contributions.

- 1) An attribution decomposition of an unrolled estimator's gain, separating the learned filter, the attention block, and unrolling itself against a matched non-unrolled control.
- 2) The consequence of a known training pathology for *attribution*: per-sample normalized error over mixed SNR concentrates the gradient at low SNR [1], so any mechanism that improves high-SNR behaviour will appear to add information, and a structural prior credited on that evidence is credited for the training's failure rather than its own contribution. We give a matched-adequacy protocol that separates the two.
- 3) Two properties of the structural contrast that a scalar hides: a per-bin sign pattern that replicates across train-

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ing designs and seeds, and a dependence of the reported sign on the averaging window.

- 4) Confirmation that the known remedy [1] transfers to unrolled magnitude-only estimation, gaining +2.171 dB at SNR ≥ 5 dB over three seeds — with a gain in every bin of every seed, including the two bins the weighting penalizes most.
- 5) A demonstration that pilot efficiency and pilot-count generalization are distinct and are conflated by a single reported curve.
- 6) A measurement of the same structural prior under three training designs, three seeds each: its advantage is +1.286 dB under the conventional loss, +0.020 dB when both arms are retrained on the regime of interest, and −0.177 dB when the loss is corrected instead.

What the control did. Under the conventional loss the prior is worth +1.286 dB at high SNR, on three seeds, positive under every statistic we tried. Retraining both arms on the high-SNR range leaves +0.020 dB, with one seed of three significantly negative. Correcting the loss instead — leaving the training range untouched — leaves −0.177 dB, negative on every seed. The quantity does not shrink by a factor; it stops keeping its sign. Two corrections that share no machinery agree, which is why we read the training design as the cause rather than an accident of either control.

II. SYSTEM MODEL AND ESTIMATOR

An N -element uniform linear array serves K users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N \times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes magnitudes only,

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2). \quad (1)$$

Each user’s channel is a sum of L_k plane waves on the array manifold, with $L_k \sim \mathcal{U}\{3, 7\}$ [7] and angles drawn uniformly on $(-90^\circ, 90^\circ)$ (Table I); this is a geometric ULA model, not a clustered one. The classical baseline is the expectation–maximization Gerchberg–Saxton iteration (EM-GS) of Cui *et al.* [5, Alg. 2], which adapts the classical Gerchberg–Saxton scheme [8] to this receiver; the unrolled estimator (“URformer”) replaces T iterations with T learned layers, each a data-consistency step followed by a learned filter, a gate, and a Transformer block across users [4].

We implement this estimator faithfully on the geometric ULA model of (1) rather than reproducing any published result; the operating point differs. Nothing here should be read as a reproduction or a validation of prior work, nor as a critique of any specific paper. The convention we examine is near-universal, and our own estimator uses it.

A. The structural intervention under test

The intervention this paper evaluates is a single denoising step applied to the per-user channel estimate inside every unrolled layer. Write $\mathcal{H}$ for the map taking a vector $\mathbf{g} \in \mathbb{C}^N$ to the $(N-p) \times (p+1)$ Hankel matrix of its sliding windows, with pencil parameter $p = \lceil N/2 \rceil - 1$, and $\mathcal{H}^\dagger$ for anti-diagonal averaging, its left inverse on Hankel matrices. With Π_r the truncation of a matrix to its r leading singular triplets, the

structured arm replaces each column $\mathbf{g}_k$ of the least-squares iterate by

$$\mathbf{g}_k \leftarrow \mathcal{H}^\dagger \Pi_r \mathcal{H} \mathbf{g}_k, \quad r = 7, \quad (2)$$

applied once per layer, at fixed rank, with no dependence on the true channel, the true path count, or σ^2 .

Four properties of (2) matter for what follows and are easy to overstate.

It is a Cadzow-style denoising step, not a projection. Truncation generally destroys the Hankel structure and re-Hankelization generally raises the rank, so one pass does not land on the intersection of the two sets; with $n_{cz} = 1$ the output is in general neither exactly Hankel-rank- r nor the nearest such matrix. Low Hankel rank also admits models beyond the unit-modulus spatial exponentials of (1), so (2) is a relaxation of the array manifold and not that manifold.

It is unconditional. There is no gate: every layer applies (2) on every trial. The estimator is therefore *not* rank-adaptive, and nothing here tests an adaptively gated structural estimator, which is a different intervention.

Its gradient is a straight-through estimator. The forward pass is exactly (2); the backward pass treats the step as the identity. The exact SVD derivative carries $1/(\sigma_i^2 - \sigma_j^2)$ terms that are numerically meaningless when singular values are close, and a fully detached projection instead severs the unrolled chain. The straight-through choice avoids both, and it is a choice: the trained network is optimized against an approximate gradient, which is itself part of the intervention being measured.

The arms are matched except for (2). Both carry the same 1,586,900 parameters, layer count, optimizer, schedule, data order, initialization and seed; (2) adds no trainable parameters.

The reference curve is genie-aided. Where a reference appears, it is an unstructured least-squares estimate formed from the *true* phase of the noiseless field. It is not a bound: it presumes information no receiver has, and being restricted to unstructured least squares it can be passed by an estimator that exploits channel structure — which we in fact observe below 5 dB in Sec. IV-A.

III. ATTRIBUTION AND THE MATCHED-ADEQUACY PROTOCOL

A. Where the gain comes from

Fig. 1 decomposes the unrolled estimator’s 3.345 dB improvement over EM-GS. The learned per-layer filter, 980 parameters, contributes 0.147 dB; the attention block contributes the remaining 3.198 dB, or most of the gain along this ablation sequence. Components can interact and the allocation depends on the order of addition, so this is one sequential decomposition rather than a unique attribution. The filter’s contribution does not grow with data — it falls slightly, $0.193 \rightarrow 0.147$ dB from 20k to 80k training samples — which is what a never-capacity-limited 980-parameter component should do.

Unrolling is doing real work independently of attention. Against a matched control that applies one Transformer block to a converged EM-GS estimate rather than unrolling, the unrolled estimator gains 0.920 dB, CI [+0.888, +0.988], winning on 86.5% of trials.

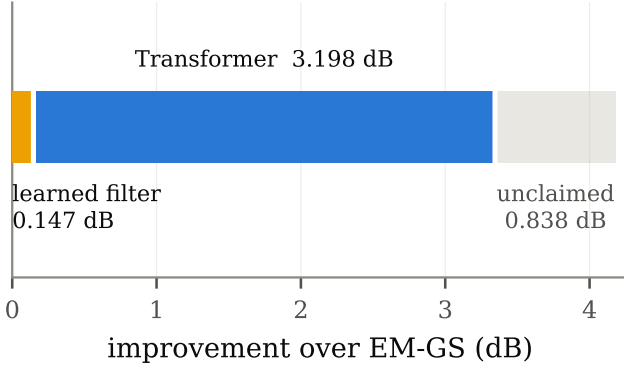


Fig. 1. Attribution of the unrolled estimator's gain over EM-GS at 80k training samples. The remainder is the gap to the genie-aided least-squares reference, which is a reference and not a bound.

B. The confound

The mechanism in this subsection is the one stated in [1]; what is new here is its magnitude in an unrolled magnitude-only estimator, and what it does to attribution.

The training loss is the per-sample normalized error $\|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 / \|\mathbf{G}\|_F^2$, averaged over a batch drawn across the whole SNR range. Because the normalized error of a low-SNR realization is orders of magnitude larger than that of a high-SNR one, this average is dominated by the low-SNR tail. Measured directly on our estimator, 89.7% of the gradient comes from trials below 5 dB, and the per-bin gradient share spans a factor of 30.0 across the range (Fig. 2, upper panel).

The concern this raises is that the high-SNR regime may be receiving too little of the optimization effort, so that *any* mechanism improving high-SNR behaviour can appear to add information. A structural prior is exactly such a mechanism. Whether the regime is genuinely underfitted is not settled by the share alone; Sec. IV-F gives a case where a well-performing estimator shows the same skew.

C. The protocol

We propose the following control, which is cheap and which we recommend be run before any structural prior is credited.

- 1) Measure the per-bin gradient share of the training loss. We define it as $g_b = \|\mathbf{g}_b\| / \sum_c \|\mathbf{g}_c\|$ with $\mathbf{g}_b$ the gradient of the loss restricted to bin b , measured on 24 batches per bin at the selected checkpoint over all trainable parameters. Because $\sum_b \|\mathbf{g}_b\| \neq \|\sum_b \mathbf{g}_b\|$, this is a normalized norm share and not an additive fraction of the net gradient, and equal share is a chosen reference rather than an optimum. A share far from uniform does not by itself prove underfitting — a small gradient can also mean a well-fitted or flat regime — so it is a signal to check regime-specific fit, not a verdict.
- 2) Retrain *both* the structured and unstructured arms with training concentrated on the regime of interest, matching data budget, schedule, seed and initialization.
- 3) Re-measure the contrast. What survives is attributable to the prior; what does not was compensating for training inadequacy.

TABLE I
CONFIGURATION.

Parameter	Value
Array elements N / users K / pilots P	32 / 3 / 20
Paths per user L_k	$\mathcal{U}\{3, 7\}$ [7]
SNR (train and test)	$\mathcal{U}[-10, 20]$ dB
Reference-to-signal ratio	10 dB
Unrolled layers T	10
Trainable parameters	1,586,900
Training samples / epochs	80,000 / 13
Test realizations	2,000 (paired)

Applying it, and reporting what it did to our own method: under the conventional loss the structural advantage over [5, 20] dB is +1.286 dB, averaged over three training seeds. Retraining both arms on [5, 20] dB alone leaves +0.020 dB. Correcting the loss instead, without narrowing the training range, leaves −0.177 dB — the prior now costs accuracy. Sec. IV-E gives the three designs with their seed spreads.

The two corrections share no machinery: one changes which channels the network sees, the other reweights an unchanged training set. They are not, however, independent in effect — both raise the relative emphasis on the high-SNR region where the contrast is evaluated, and neither isolates undertraining from objective alignment, sample allocation or optimization geometry. Their agreement therefore supports training-design sensitivity; it does not by itself identify a mechanism.

Scope of this negative result. It holds at $N = 32$, $K = 3$, this data budget, and the two pilot counts tested. It is not a general claim that structural priors cannot help unrolled estimators. Indeed the same prior applied to the *classical* estimator, where no training adequacy is at stake, gives a large and stable gain, and under a distribution shift in path richness the structured network degrades least (+0.48 dB against +0.73 dB). That is a smaller observed degradation in one shift test, not a demonstrated robustness advantage: we do not report absolute shifted error for both arms, and a method can start worse and deteriorate less. It is the in-distribution accuracy claim that does not survive the control.

IV. NUMERICAL RESULTS

Table I lists the configuration. The reported statistic is the paired per-trial median within an SNR bin, with bootstrap 95% confidence intervals over 2000 paired test realizations.

A. The known remedy transfers to this setting

The confound is a property of the loss, and the remedy for it is already known [1]. This subsection reports what it is worth here. That is a confirmation, not a contribution, and it is placed first only because the rest of the section is easier to read once the loss is fixed.

We weight each sample by a static per-bin factor $w(b) = c/m(b)$, where $m(b)$ is the mean per-sample normalized error in bin b measured once before training, with c set so the weights have unit mean. This requires no schedule, no tuning, and no change to the architecture.

The differences from [1] are worth stating exactly. Their weights are recomputed each epoch from inverse accumulated losses, with a stability constant and a normalization at the centre of their SNR grid; ours are computed once and never change. They also note at the end of their Section 4 that one could instead deweight using the loss of a fixed baseline such as a classical system, rather than the adaptive scheme they implement — a fixed-weight variant they raise but do not implement. Ours is fixed in that sense, with the further difference that the weights come from the network’s own pre-training error and not from a separate classical baseline.

It works as designed. The realized per-bin gradient share spans a factor of 1.9 after balancing against 30.0 before, and the share below 5 dB falls from 0.897 to 0.427 against an ideal of 0.5 — a slight over-correction.

The estimation result is better than we predicted, and it is measured on three seeds rather than one. Against identically-trained unbalanced baselines, balancing gains +1.902, +2.242 and +2.369 dB at $\text{SNR} \geq 5$ dB, a mean of +2.171 dB. Each seed’s bootstrap 95% confidence interval *over test realizations* excludes zero. The spread *over seeds* is a separate quantity and is reported as one: its standard deviation is 0.241 dB, on three seeds and therefore two degrees of freedom, so it is indicative rather than precise. A two-sided 95% Student-*t* interval for the seed mean is [+1.572, +2.771] dB, which excludes zero; with three runs the normality that interval assumes cannot itself be checked.

We had pre-registered a low-SNR *cost* of -0.10 to -0.40 dB on the reasoning that the lowest bin’s weight falls about eighteenfold. There was no cost. On the seed for which the prediction was registered the low-SNR change is +0.135 dB, CI over test realizations [+0.101, +0.172]; across the three seeds the two lowest bins gain +0.052 and +0.202 dB on average. Downweighting the bins that dominated the gradient did not harm them, because they were over-served rather than merely well-served.

On the seed for which the prediction was registered, the gap to the genie-aided reference at $\text{SNR} \geq 5$ dB falls from +2.897 to +1.000 dB; that difference is the first seed’s improvement and not the three-seed mean. Below 5 dB the balanced estimator passes that reference by up to 1.99 dB, which is consistent rather than anomalous: the reference is restricted to unstructured least squares, and an estimator that has learned the channel geometry is not so restricted.

B. The correction improves the regime it penalizes

The obvious objection to a reweighting is that it trades one regime for another: whatever the high-SNR bins gain, the down-weighted low-SNR bins must have paid. Here they did not, and the check is exhaustive rather than pooled.

Fig. 3 plots every SNR bin for every seed. **All eighteen bin-by-seed cells favour the balanced loss**, and seventeen of the eighteen have a bootstrap 95% confidence interval over test realizations that excludes zero; the single exception is seed 1’s $[-5, 0)$ dB bin at +0.044 dB, CI $[-0.011, +0.105]$. The three-seed means rise monotonically across the range, from +0.052 dB in $[-10, -5)$ to +3.120 dB in $[15, 20)$.

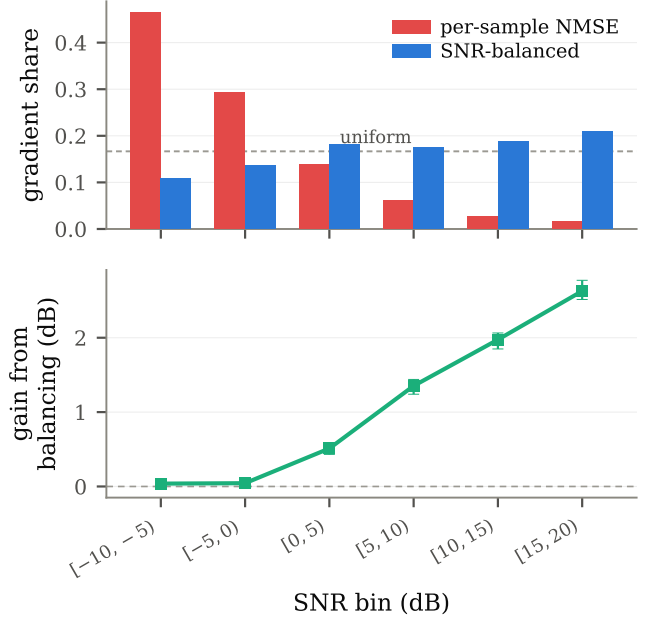


Fig. 2. Cause and effect. Upper: per-bin gradient share under the per-sample normalized loss and under SNR balancing. Lower: the resulting per-bin gain, paired per-trial median with bootstrap 95% CI over test realizations, on the seed for which the prediction was registered.

The two lowest bins are the ones that matter for the objection, because the weighting divides their contribution by about 18 and 9 relative to unit mean. They gain anyway. The correction is therefore not a transfer between regimes: it improves the quantity it was weighted away from.

This differs from what the prior work reports. Wiesmayr *et al.* [1] do not report improvement in the regime their deweighting penalizes; where they examine it they report the trade. In their narrow-range experiment, deweighted training holds a small advantage at high SNR while naive training keeps a smaller one at the low end of the waterfall, and their replication on a simpler system repeats the pattern. Their wide-range result is a 0.54 dB gain at 1% block error rate, which is a high-SNR operating point; no per-bin low-SNR improvement is claimed.

Three differences must be kept in view before that contrast is read as a disagreement. *Metric*: they measure bit and block error rate on a coded MIMO-OFDM system, we measure normalized channel-estimation error, so “improvement in the down-weighted bins” is not the same quantity. *Mechanism*: their weights are recomputed each epoch from inverse accumulated losses with a stability constant and a normalization at the grid centre; ours come from one pass before training. *Size*: their gain is 0.54 dB, which they describe as modest but essentially free since it comes from the loss rather than a larger receiver or more data. Because the metrics differ, their number and ours are not directly comparable and we do not compare them.

We offer a reading of why, as a hypothesis and not as a result: under the per-sample loss the low-SNR tail may dominate to the point that the network is underfitted *everywhere*,

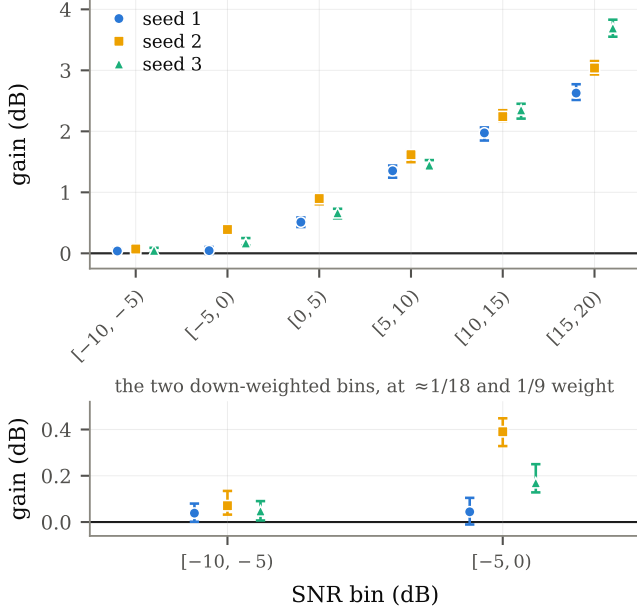


Fig. 3. Every bin, every seed. The gain from SNR balancing, paired per-trial median with bootstrap 95% CI over test realizations, for each of three training seeds. All eighteen cells are positive, including the two lowest bins, whose weights the balancing divides by about 18 and 9.

TABLE II

PER-BIN STRUCTURAL GAIN Δ_H , THREE-SEED MEAN, POSITIVE = THE PRIOR IS BETTER. ALL EIGHTEEN UNDERLYING Δ_H CELLS OF THE MIXED DESIGN — THREE SEEDS BY SIX BINS — HAVE A BOOTSTRAP 95% CI OVER TEST REALIZATIONS EXCLUDING ZERO. THESE ARE A DIFFERENT EIGHTEEN FROM THOSE OF FIG. 3, WHICH ARE THE GAIN FROM BALANCING THE LOSS.

SNR bin (dB)	mixed	SD over seeds	focused	balanced
$[-10, -5)$	-0.120	0.036	—	-0.233
$[-5, 0)$	-0.400	0.103	—	-0.703
$[0, 5)$	-0.293	0.157	—	-0.944
$[5, 10)$	+0.457	0.052	-0.173	-0.692
$[10, 15)$	+1.403	0.089	+0.067	-0.241
$[15, 20)$	+2.290	0.082	+0.259	+0.567

in which case rebalancing stops wasting capacity rather than relocating it. We have not tested that. The measurement is that the reweighting improved every bin-by-seed point estimate; the mechanism is a conjecture, and the two should not be read together. Note also that these are eighteen individual descriptive intervals, not simultaneous inference, and that all are medians in decibels — a positive median in every bin does not guarantee improved expected squared error.

C. The structural contrast is not a scalar: a per-bin sign pattern

Reported as one number, the structural gain Δ_H is a weighted average of two opposing effects. Per bin, the prior costs accuracy at low SNR and buys it at high SNR, and the sign flips exactly once.

Two things about Table II are worth separating.

The ordering replicates and the crossing does not. Every arm we have measured — three seeds of mixed-SNR training,

three of focused training, and an earlier arm under the balanced loss — places every negative bin before every positive one. On all three mixed seeds the sequence is exactly $- - - + + +$ with all eighteen intervals excluding zero. But the bin at which the sign turns moves by up to two positions between arms. The ordering is the property that survives; the crossing SNR is not, and neither is any scalar summary that depends on where it falls.

The magnitude falls under matched adequacy while the ordering does not. Comparing the two columns, the two highest bins shrink by factors of about 21 and 9, and the $[5, 10)$ bin changes sign rather than shrinking, from +0.457 to -0.173 dB.

D. The averaging window decides the reported sign

Because the per-bin contributions have opposite signs, the number one reports depends on the window the average is taken over — both which bins are inside it and how trials are combined within it. Across three seeds of each design:

- mixed-SNR, restricted to $\text{SNR} \geq 5$: +1.209, +1.351, +1.298 dB by paired per-trial median; +0.723, +0.856, +0.901 dB by ratio of sums. Positive under both.
- mixed-SNR, pooled over the whole range: +0.129, +0.225, +0.196 dB by median; -0.214, -0.146, -0.141 dB by ratio of sums. *Sign disagreement*.
- focused training, $\text{SNR} \geq 5$: +0.078, +0.057, -0.076 dB by median; -0.059, -0.077, -0.234 dB by ratio of sums. Negative under ratio of sums on every seed.

The reason is measurable and we measured it. The paired median weights every trial equally; the ratio of sums weights each trial by its own squared error. Sorting trials by the no-prior arm's error, the top two deciles carry 51.2% of the total and contribute -0.134 dB, while the remaining eight contribute essentially nothing. That is not a conditioning effect separate from SNR: within an SNR bin the rank correlation between the baseline error and Δ_H is -0.009, -0.046 and +0.147, against -0.161 overall and +0.198 against SNR itself. The sum of squared errors is dominated by low-SNR realizations, which is exactly where Table II says the prior costs accuracy.

It is not the mechanism that produces the same symptom in the classical estimator, where a selection rule abstains on most trials and pins the median at zero. The unrolled arm has no such rule: the projection is applied unconditionally, and it moves the estimate on every one of the 10 layers of every one of 2000 trials on every seed, with no trial anywhere near inactive.

E. The structural prior under matched adequacy

Table III is the paper's central measurement. The prior, the architecture, the data budget, the pilot count and the test set are identical across its three rows. What differs is only how training distributed its effort.

Read row by row, the three designs give +1.286 dB, a value near zero, and a negative value. Read as *paired differences* — the same seed on both sides of each comparison —

TABLE III
THE STRUCTURAL GAIN Δ_H UNDER THREE TRAINING DESIGNS, THREE SEEDS EACH, ALL ARMS SPECTRALLY INITIALIZED. POSITIVE = THE PRIOR IS BETTER. INTERVALS ARE TWO-SIDED 95% STUDENT- t INTERVALS FOR THE SEED MEAN ON TWO DEGREES OF FREEDOM; THEY DESCRIBE TRAINING-SEED VARIABILITY, NOT TEST-SET UNCERTAINTY. RoS IS THE SIGN OF THE RATIO-OF-SUMS STATISTIC PER SEED.

Training design	Δ_H	SD _{seed}	95% CI (seeds)	RoS
Conventional loss	+1.286	0.072	[+1.107, +1.465]	+ + +
Focused [5, 20]	+0.020	0.083	[-0.187, +0.226]	- - -
Balanced loss	-0.177	0.150	[-0.549, +0.196]	- - -
<i>Paired per-seed change from conventional training</i>				
Focused — conventional	-1.267	0.124	[-1.574, -0.959]	
Balanced — conventional	-1.463	0.179	[-1.908, -1.018]	

the change from conventional training is -1.267 dB under focused retraining and -1.463 dB under the balanced loss, with intervals excluding zero in both cases ($p = 0.003$ and $p = 0.005$, paired t on two degrees of freedom).

The change is what this experiment establishes; the endpoints are not. On three seeds the focused and balanced contrasts are each consistent with zero: their intervals are $[-0.187, +0.226]$ and $[-0.549, +0.196]$. We therefore do not claim that the structural advantage is absent under focused training, nor that it is reliably negative under the balanced loss. What the paired analysis does support, and support strongly, is that the measured advantage moves by more than a decibel when only the training design changes.

The two corrections share no machinery: one changes which channels the network sees, the other reweights an unchanged training set. Both nonetheless raise the relative emphasis on the high-SNR region where the contrast is evaluated, and neither isolates undertraining from objective alignment, sample allocation or optimization geometry. Their agreement establishes training-design sensitivity; it does not identify a mechanism.

What we do not claim. A structural prior can deliver a real finite-budget benefit through optimization or regularization even when that benefit disappears under a different training design; a conditional practical benefit is not the same thing as a training-independent architectural advantage, and this experiment separates the two rather than refuting the first. We have not measured Δ_H under the log-domain loss of Sec. IV-F: that needs a prior-carrying arm trained under that loss, which we did not train. Three designs are reported, not four. Nor is this a claim about structural priors in general — it holds at $N = 32$, $K = 3$, this data budget and these pilot counts, for the specific operator of Sec. II-A, applied unconditionally at every layer. An adaptively gated structural estimator is a different intervention and is not tested here.

F. A simpler loss, with no resolved difference

Per-bin reweighting needs bin edges, a measurement pass before training to estimate the weights, and a choice of binning. Taking the per-sample error in decibels before averaging needs none of them: the logarithm divides out the scale factor exactly, per sample. It is the obvious simplification, so we tested it on three seeds.

Against the bin-reweighted arm it gains $+0.121$ dB at $\text{SNR} \geq 5$ dB, with an across-seed SD of 0.131, so the difference is not separated from zero; against the conventional arm both fixes gain about the same, $+2.290$ dB for the log loss. A two-sided 95% Student- t interval for the seed mean is $[-0.204, +0.446]$ dB, which lies inside the ± 0.50 dB equivalence margin registered before these runs, so the pre-registered equivalence criterion is met. **Stated without reference to that margin, the three-seed comparison simply does not resolve a difference**, and we prefer that wording where no margin is in view. Either way the log-domain loss buys no measured improvement over per-bin reweighting, and we report it because a reader will ask.

It is not, however, better balanced, and here a registered prediction of ours was wrong in the direction of the error: we predicted the log loss would over-correct *less* than per-bin reweighting, and it over-corrects more. Its realized share of gradient below 5 dB is 0.317–0.329 against an ideal of 0.5, where per-bin reweighting reaches 0.427; its per-bin span is 3.6 against the reweighted arm’s 1.9. Dividing the scale out exactly does not equalize the gradient, because the derivative of a logarithm is itself inversely proportional to the error.

The skew is a property of the loss, not of the weights — and this limits our own diagnostic. The same log-trained networks, scored under the conventional objective, put 0.906–0.911 of the gradient norm below 5 dB with a span above 40. These networks perform well; they are not undertrained in any sense we can detect. A skewed gradient share therefore cannot certify training inadequacy, and the first step of the protocol in Sec. III should be read as a prompt to check regime-specific fit, not as a test that returns a verdict.

G. Pilot efficiency is not pilot-count generalization

A pilot-count curve for a trained estimator can mean two different things: one model trained at some P_0 and evaluated at each P (generalization), or a model trained at each P (efficiency). These are not close. Trained at $P = 10$, the estimator beats the $P = 20$ model evaluated there by $+2.231$ dB pooled, every bin’s CI excluding zero. At $P = 35$ the corresponding margin is $+0.635$ dB, so the model generalizes upward in pilot count far better than downward.

At 5 dB the matched-trained estimator reaches -7.52 dB at $P = 10$, within 0.47 dB of the genie-aided reference, while the $P = 20$ model evaluated there is 2.60 dB short of it. The clearest single indication that the curve is tied to its training condition is that the $P = 20$ model is no better at $P = 25$ (-11.13 dB) than at $P = 20$ (-11.27 dB). That difference is 0.14 dB and we report it without paired uncertainty, so it indicates the absence of the expected improvement rather than established harm from additional measurements. Fig. 4 shows all three curves together.

This matters for how such curves are reported. The pilot sweep of [4] is described as “the NMSE performance versus the number of pilot transmissions P , evaluated at a fixed SNR of 5 dB”, and as “crucial for assessing the pilot efficiency of each algorithm”; the text does not state whether the networks were retrained per pilot count. We report that as a fact about

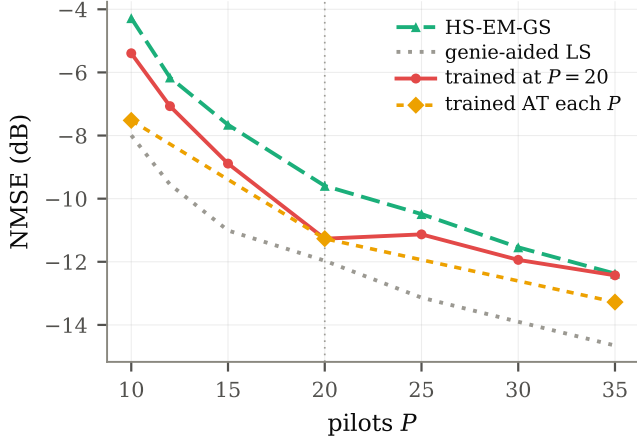


Fig. 4. Pilot efficiency against pilot-count generalization at 5 dB, on identical realizations. The dashed diamonds are separate networks trained at each P ; the solid curve is one network trained at $P = 20$.

what the paper specifies, and as motivation for stating the distinction explicitly — not as a criticism of that work, whose convention is the ordinary one and which our own earlier curves shared.

V. CONCLUSION

A training loss averaged over a wide SNR range is dominated by its low-SNR tail [1]; here 89.7% of the measured gradient norm falls below 5 dB. The consequence we draw is about evidence rather than performance: a mechanism that improves high-SNR behaviour can appear to add information it need not have supplied, so an uncontrolled comparison does not establish that a structural prior supplied it. We recommend the matched-adequacy control be run before a structural prior is credited, and note that it cost us our own result first.

Two properties of the structural contrast survive that control and a single number does not report either: the per-bin sign pattern, which replicates across training designs and seeds while the SNR at which it crosses does not, and the dependence of the reported sign on the averaging window. Applying the known remedy [1] to this setting gains +2.171 dB at $\text{SNR} \geq 5$ dB across three seeds and improves every one of eighteen bin-by-seed cells, including the two the weighting itself penalizes most.

Applied to our own Hankel-structured estimator, the control moves a +1.286 dB advantage by −1.27 dB under focused retraining and by −1.46 dB under a corrected loss, measured as paired per-seed differences with intervals excluding zero — three designs, three seeds each, nothing changed but how training distributed its effort. The endpoints themselves are not separated from zero on three seeds, and we do not claim them as absent or negative.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They also cover one architecture, so we cannot separate what is a property of training design in general from what is a property of this network; [1] repeats its whole experiment on a second, simpler system precisely to rule that out, and the corresponding check

here has not been run. They cover the geometric ULA model of (1) at $N = 32$, $K = 3$, one data budget and the pilot counts tested; the negative result about structural priors is scoped to those conditions and is not a general claim. We establish no fundamental bound: the reference curve used throughout is genie-aided, presumes phase no receiver observes, and is passed by our own estimator at low SNR.

Data and Pre-Registration Records

Every prediction called pre-registered above exists as a dated record written before the corresponding runs started. Each record fixes the threshold, the falsifier and the rule that would decide the outcome, and each was scored held or failed after the runs finished. The records, the scoring scripts and the per-run result files accompany this manuscript. Two of the predictions in this paper failed and are reported as failures in the text: the predicted low-SNR cost of balancing, and the prediction that the log-domain loss would correct the gradient share less than per-bin reweighting does.

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